

Computational surveillance of Colombian public pharmaceutical procurement using public administrative data: a reproducible analysis of a closed 2020-2025 cohort

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Abstract

Introduction. BigLoI monitors Colombian public pharmaceutical procurement from 2015 onward. For this manuscript, the closed analytical cohort comprised 162,271 pharmaceutical contracts from 2020 to 2025, avoiding year-to-year comparisons with incomplete 2026 records.

Objective. To describe the design, implementation, and findings of a reproducible computational infrastructure for surveillance of Colombian public pharmaceutical procurement.

Methods. Public APIs from SECOP-II, INVIMA, and SISMED were integrated into a reproducible architecture combining PostgreSQL, statistical analysis, and a public-facing observatory. A Z-score engine was implemented to flag contracts with atypical total values within therapeutic categories. As a clearly secondary technical-feasibility module, smart-contract automation of a payment workflow was tested on the Sepolia testnet only to verify predefined digital state transitions under simulated conditions. The closed analytical cohort included 162,271 pharmaceutical contracts from 2020 to 2025, while post-2025 records remained available only for live platform monitoring. Monetary results were reported primarily in COP; secondary USD equivalents were included only as approximate interpretive references, with detailed conversions relegated to Supplementary Table S1.

Results. Among 147,020 contracts analyzed in categories with at least 10 observations, 685 contracts (0.47%) triggered a statistical alert with absolute Z-score greater than or equal to 1.5 sigma. The alert rate increased from 0.31% in 2021 to 1.38% in 2025. Antibiotics showed the highest category-level maximum Z-score (8.32). The top 3% of suppliers concentrated 85.9% of total contracted value. In the secondary automation module, the Sepolia prototype confirmed only that predefined digital state transitions could be executed under test conditions and supported only an illustrative financial-savings scenario of COP 224,000-330,000 million per year (approximately USD 56.0-82.5 million at a reference rate of COP 4,000/USD) under favorable assumptions about administrative-delay reduction; it did not measure real institutional processing times or realized savings.

Conclusions. A reproducible national-scale computational infrastructure identified atypical procurement patterns and documented marked market concentration in Colombian public pharmaceutical procurement. These findings are relevant for auditing, public health policy, and health data governance. The blockchain module should be interpreted strictly as a complementary technical proof of concept and not as operational evidence on real-world payment performance, savings, or implementation readiness.

Keywords: public pharmaceutical procurement; SECOP-II; procurement surveillance; market concentration; anomaly detection; artificial intelligence; blockchain; SISMED; INVIMA; Colombia.

1. Introduction

Public procurement of medicines in Colombia is monitored in BigLoI through a broad SECOP-II universe that now exceeds 272,000 contracts from 2015 onward. For the purposes of this manuscript, the closed indexed analytical cohort corresponds to 162,271 pharmaceutical contracts from 2020 to 2025. Despite this scale, the system lacks reproducible real-time surveillance tools to detect atypical procurement patterns, market concentration, or potential supply risks.

The World Health Organization has estimated that 10% to 25% of global health expenditure is lost to corruption, inefficiency, and administrative errors (2). In Colombia, official audits have documented high-cost medicine overpricing, prolonged payment delays, and recurring episodes of shortages (3,4).

Integration of public APIs makes it possible to build procurement-surveillance platforms without access to confidential information. Z-score analysis is a practical approach for detecting atypical patterns in large procurement datasets (6), while smart contracts and semantic retrieval expand the range of possible automation and evidence-access tools (7,8). Initiatives such as ProZorro, OCDS, and OLAF illustrate the feasibility of these approaches in public-procurement transparency (9-11).

BigLoI (Business Intelligence for Government Logistics and Operations Intelligence) was developed to address this gap. This article documents its methodology and findings as a contribution to the debate on computational tools for governance of the Colombian health system.

2. Methods

2.1 Study design

This was a technological-development study with longitudinal observational analysis of public procurement data. Only openly accessible Colombian public data sources were used. No clinical data or patient-level personal data were included. Under Colombian Resolution 8430 of 1993, the study is classified as no-risk research.

2.2 Data sources

SECOP-II (Electronic Public Procurement System). The public Socrata API from datos.gov.co was used for automated download of pharmaceutical procurement contracts. Inclusion required the presence of at least one of 19 pharmaceutical terms in the contract-object description. Obvious false positives were excluded. The platform monitoring window extends from January 2015 onward, but the closed analytical cohort used in this manuscript was restricted to complete calendar years 2020-2025.

SECOP-II monitored universe available to the platform: 272,814 unique contracts; COP 42 trillion; 163,135 suppliers; 37 regional codes

Recovered local PostgreSQL source used to reproduce the manuscript cohort: 956,157

total SECOP-II records; 165,549 pharmaceutical records including live 2026 monitoring rows; 162,271 pharmaceutical contracts in the closed 2020-2025 analytical cohort

To avoid comparing incomplete calendar periods, all year-by-year descriptive analyses in the manuscript were closed at 31 December 2025. The platform continues to ingest newer records for operational monitoring, but post-2025 records were excluded from the formal analytical cohort reported here.

All primary monetary values are reported in COP. When approximate USD equivalents are provided in parentheses, they are secondary interpretive aids calculated using a round reference exchange rate of COP 4,000 per USD, close to late-2025 market levels. No analyses were performed in USD. Detailed COP/USD reference conversions for the principal manuscript figures are provided separately in Supplementary Table S1.

INVIMA (National Food and Drug Surveillance Institute). A total of 9,838 records from the national sanitary registry were downloaded through the Socrata API. Variables included brand name, active ingredient, marketing authorization holder, sanitary registration number, ATC code, dosage form, route of administration, relevant dates, and registry status. All loaded records were current.

SISMED (Medicine Price Information System, Ministry of Health). We processed 44,038 reference-price records covering 1,759 unique ATC codes, aggregated by dosage form, distribution channel, and reference year (2017-2019). These data were used as contextual market references and as complementary input for interpreting atypical contract-value findings.

2.3 BigLoI technical architecture

The platform was implemented as a monorepo with a seven-layer architecture (Table 2).

Table 2. Technical architecture of the BigLoI platform for surveillance of public pharmaceutical procurement

Layer	Main technology	Function
Collection	Python; Socrata API	Incremental ingestion of SECOP-II, INVIMA, and SISMED
Storage	PostgreSQL; Pinecone; MongoDB	Relational, vector, and unstructured data storage
Processing	FastAPI; pandas	Cleaning, normalization, ABC classification, Z-score engine
Generative AI	Claude 3.5 Sonnet; GPT-4o	Hybrid semantic plus TF-IDF retrieval-augmented generation
Machine learning	scikit-learn	Demand prediction ($R^2 > 0.85$), k-means (6 groups), PCA
Smart contracts	Solidity; Chainlink CRE; Sepolia	Payment automation across five states; invoice NFTs
Visualization	React/TypeScript	Public observatory with time series, maps, and alerts

2.4 Anomaly-detection engine based on Z-scores

For each pharmaceutical contract classified into a therapeutic category, we calculated the standardized deviation of its total contract value relative to the distribution of observed values within the same category. The analysis was restricted to categories with at least 10 contracts to avoid unstable estimates:

$$Z_i = \frac{V_i - \bar{V}_{cat}}{\sigma_{cat}}$$

where Z_i is the anomaly score for contract i , V_i is the contract’s total value, $\bar{V}_{cat}$ is the mean contract value within the same therapeutic category, and σ_{cat} is the standard deviation of that category-specific distribution. This approach detects contracts with atypical total values relative to their category; by itself, it does not establish unit-price overpricing against a regulatory benchmark.

Alert levels were classified as follows:

Level	Criterion	Interpretation
Critical	Absolute Z-score greater than or equal to 3.0 sigma	Extremely atypical contract value within category
High	Absolute Z-score from 2.0 to less than 3.0 sigma	Clearly atypical contract value
Moderate	Absolute Z-score from 1.5 to less than 2.0 sigma	Moderately atypical contract value
Normal	Absolute Z-score below 1.5 sigma	Within the expected range for category

2.5 Market-concentration analysis

To characterize the competitive structure of the pharmaceutical procurement market, we calculated descriptive concentration indicators and estimated the proportion of total contracted value accumulated by leading suppliers. Supplier concentration was calculated over positive-value contracts in the closed 2020-2025 pharmaceutical cohort. The Herfindahl-Hirschman Index (HHI) was calculated as the sum of squared supplier shares of total contracted value and is reported on the conventional 0-10,000 scale. Bipartite graphs of buyers and suppliers were constructed to identify recurrent contracting patterns.

2.6 Complementary technical-feasibility analysis of a payment workflow using smart contracts

We implemented a five-state simulator on the Sepolia testnet solely to verify that predefined digital states of a hypothetical pharmaceutical payment workflow could be executed in sequence under controlled test conditions. Comparison with current practice was performed only at the level of temporal order of magnitude using an aggregated institutional workflow as reference and was not designed as an empirical time-and-motion study. For the operational segment, local Servicio Farmaceutico evidence on need identification, ordering, supply, reception, registration, and control was considered. For the administrative-financial segment, a general sequence of invoice filing, accounting causation, and payment compatible with public hospitals was used. The 90-day median

and 60-180 day range were retained as aggregated institutional references and not as a step-by-step operational equivalence between the full hospital workflow and the digital prototype states. The prototype reference scenario was approximately 30 hours and should be read only as a test-condition benchmark for the simulator.

The illustrative projected savings were estimated as a scenario exercise, not as an empirical economic evaluation, by applying an average sector financial cost of 2% per month to annual contracted value under a hypothetical scenario of substantial reduction in administrative delays:

$$Annual\ savings = V_{annual} \times r_{monthly} \times \frac{\Delta t_{days}}{30}$$

where V_{annual} is approximately COP 3.7 trillion, $r_{monthly} = 2\%$, and Δt represents the theoretical reduction in the digital administrative component of the payment cycle.

2.7 Statistical analysis

Descriptive statistics were used throughout. Log-normality of contract values was assessed with the Kolmogorov-Smirnov test. Market concentration was described using participation and accumulation indicators. The significance threshold was set at $\alpha = 0.05$.

3. Results

3.1 Characteristics of the data corpus

Table 1. Summary of the public pharmaceutical procurement corpus and the closed analytical cohort used in this manuscript

Variable	Value
Total SECOP-II contracts (API universe)	272,814
Pharmaceutical contracts indexed in PostgreSQL (closed 2020-2025 cohort)	162,271
Total contracts in recovered local PostgreSQL source	956,157
Total contracted value (closed pharmaceutical cohort)	COP 17.07 trillion (approx. USD 4.27 billion)
Total contracted value (all sectors recovered local source)	COP 294.7 trillion
Total contracted value (API universe)	COP 42.00 trillion
Mean value per pharmaceutical contract	COP 105 million
Unique pharmaceutical suppliers in closed cohort	50,460
Regions covered	36-37
Period (closed pharmaceutical cohort)	2020 to 2025
Platform monitoring window	January 2015 onward
Processed INVIMA records	9,838 (all current)

Variable	Value
SISMED reference prices	44,038 records; 1,759 unique ATC codes (2017-2019)
Indexed RAG documents	9,336

Pharmaceutical contract value followed a log-normal distribution (Kolmogorov-Smirnov $D = 0.04$; $p < 0.001$). The top 10% of contracts by value accounted for 78% of total contracted value.

Annual evolution of indexed pharmaceutical contracts (closed cohort, 2020-2025):

Year	Contracts	Value (COP billions)
2020	35,330	3,302.6
2021	53,832	2,856.7
2022	30,991	2,648.4
2023	10,705	1,603.1
2024	9,380	1,137.5
2025	22,033	5,519.3

The peak in contract counts in 2021 is consistent with the COVID-19 response period, whereas the value peak in 2025 coincided with increased participation of high-cost oncology procurement. Post-2025 records remain available in the live platform but were excluded from the closed annual comparisons reported here.

Distribution by therapeutic category:

Category	Contracts	Value (COP billions)	Mean value (COP millions)
General pharmaceutical	82,303	9,990.3	121.4
Medical device	15,943	2,957.0	185.5
Vaccine	46,130	905.3	19.6
Oncology	403	422.8	1,049.2
Medical supply	2,312	201.6	87.2
Antibiotic	219	21.7	98.9
Diabetes	281	20.8	74.0
Analgesic	42	8.5	203.0
Antiviral	37	0.9	24.6

Oncology contracts showed the highest mean contract value (COP 1,049.2 million per contract).

3.2 Geographic distribution

The leading regions by pharmaceutical procurement value were as follows:

Region	Contracts	Value (COP billions)	Approx. value (USD millions)	% of total
Bogota D.C. (SECOP code 1)	6,223	4,877.2	1,219.3	28.4%
Bogota D.C. (SECOP code 2)	12,093	4,788.4	1,197.1	27.9%
Bogota D.C. (combined)	18,316	9,665.6	2,416.4	56.3%
Antioquia	48,039	1,316.2	329.1	7.7%
Valle del Cauca	5,888	706.7	176.7	4.1%
Boyaca	9,195	633.0	158.3	3.7%
Huila	4,212	472.8	118.2	2.8%
Tolima	3,816	431.6	107.9	2.5%
Santander	4,319	418.8	104.7	2.4%
Cauca	2,023	395.6	98.9	2.3%
Atlantico	5,087	351.4	87.9	2.0%

Bogota D.C. accounted for 56.3% of national contracted value, equivalent to approximately USD 2.42 billion at the reference exchange rate, followed by Antioquia (7.7%) and Valle del Cauca (4.1%).

Methodological note. Bogota D.C. appears under two regional codes in SECOP-II. Coverage of 36-37 regions exceeds the 32 official departments because SECOP-II includes distinct codings for districts and other territorial entities.

3.3 Contract-value anomalies detected by the Z-score engine

Methodological note. The implemented Z-score engine measures deviation of total contract value relative to the distribution within each therapeutic category. Direct comparison against SISMED unit prices would require structured line-item detail, which is not consistently available in SECOP-II.

Among 147,020 analyzed contracts in categories with at least 10 observations, the engine identified the following:

Alert level	Criterion	Contracts	% of total
Critical	Absolute Z-score greater than or equal to 3.0 sigma	303	0.21%
High	Absolute Z-score from 2.0 to less than 3.0 sigma	225	0.15%
Moderate	Absolute Z-score from 1.5 to less than 2.0 sigma	157	0.11%
Total with alert	Absolute Z-score greater than or equal to 1.5 sigma	685	0.47%

Results by therapeutic category:

Category	Contracts	% with alert	Zmax	Max/mean ratio
Analgesic	42	9.5%	4.22	9.3x
Diabetes	281	8.2%	6.58	15.3x
Antiviral	37	5.4%	4.83	14.2x
Antibiotic	219	3.7%	8.32	24.1x
Medical supply	2,312	2.6%	22.45	86.6x
Oncology	403	0.7%	19.34	139.0x

Antibiotics showed the highest category-level maximum Z-score, whereas analgesics showed the highest proportion of contracts with alerts among the categories displayed in Figure 3.

Temporal evolution of anomaly rates (2020-2025):

Year	Contracts analyzed	Contracts with alert	Rate
2020	35,326	121	0.34%
2021	53,827	165	0.31%
2022	30,990	130	0.42%
2023	10,705	89	0.83%
2024	9,380	86	0.92%
2025	6,792	94	1.38%

The statistical-alert rate increased from 0.31% in 2021 to 1.38% in 2025. Causal interpretation of this pattern requires additional analysis.

Geographic distribution of anomalies.

Departments with the highest rates of statistically flagged contracts were Bogota D.C. (3.80%), Norte de Santander (1.92%), Valle del Cauca (1.21%), and Tolima (1.00%). This distribution indicates heterogeneity in contract values, not causal evidence of corruption.

3.4 Market concentration

The concentration analysis of Colombian pharmaceutical procurement in the closed analytical cohort showed 50,460 unique supplier names. For value-share concentration metrics, 162,151 positive-value contracts and 50,355 suppliers with non-empty normalized labels were included. The top 10 suppliers concentrated 28.77% of total contracted value and the top 3% of suppliers (1,511 of 50,355) concentrated 85.87%. The leading supplier, VECOL SA, accounted for 5.61% of total value. The HHI was 119.26 on the 0-10,000 scale. The detailed distribution is shown in Figure 4.

This pattern indicates strong cumulative concentration among leading suppliers, while the HHI remains low under conventional market-concentration thresholds because the contracted value is distributed across a large long-tail supplier base. The bipartite graph identified 847 recurrent entity-supplier pairs with at least five consecutive contracts.

3.5 Smart-contract payment workflow: complementary technical proof of concept

Tests in the Sepolia-based BigLoI plus Chainlink CRE simulator verified that the prototype could execute predefined transitions between digital states. Reported times are investigator-defined reference times used for functional validation of the prototype and do not represent observed measurements of the full real-world logistics and administrative process. SECOP-II was used as a contractual reference point rather than as an operational stage of hospital workflow, and no hospital payment system was instrumented or experimentally compared.

Functional stage	Prototype time
Validated contract or enabled order to medicines dispatched	About 24 hours
Medicines dispatched to delivery verified	About 4 hours
Delivery verified to CRE registered	About 1 hour
CRE registered to payment released	About 1 hour
Full digital cycle of the prototype	About 30 hours

The current cycle documented in literature and institutional reports is 60-180 days, with a 90-day median. This comparison should be interpreted only as a heuristic contrast in order of magnitude for the digital administrative component, not as a direct equivalence to the full hospital process or as evidence that the prototype would reproduce those reductions in practice.

Illustrative financial-savings scenario.

Average annual pharmaceutical procurement flow across the closed 2020-2025 cohort was approximately COP 3.4 trillion (about USD 853 million at the reference exchange rate). Under a theoretical scenario of substantial reduction in digital administrative delays, and applying a monthly financial cost of 2% to the avoidable payment-cycle component, projected annual savings were estimated as COP 224,000 million per year (approximately USD 56.0 million). Using 2025 annual flow, the illustrative potential rises to COP 330,000 million per year (approximately USD 82.5 million). These values are scenario outputs only and should not be interpreted as observed savings, budget impact estimates, or implementation-ready business cases.

3.6 Complementary modules: machine learning and semantic retrieval

The complementary machine-learning and semantic-retrieval modules showed operational feasibility, but their outputs should be interpreted as exploratory within the scope of this paper.

4. Discussion

4.1 Scale and relevance of the documented problem

The corpus of 162,271 pharmaceutical contracts from 2020 to 2025 represents, to our knowledge, the largest validated closed longitudinal series of public pharmaceutical procurement described for Colombia. The log-normal distribution of contract values, the concentration of 78% of total value in the top decile, and the HHI of 119.26 are consistent with procurement markets characterized by a small group of high-value suppliers and a large long-tail supplier base (14).

The increase in statistical-alert rates from 0.31% in 2021 to 1.38% in 2025 suggests increasing heterogeneity in contracted values during the study period. This pattern warrants dedicated investigation with adjustment for potential confounders such as pharmaceutical inflation and the introduction of new high-cost categories.

4.2 Z-scores as a surveillance instrument

The Z-score methodology offers advantages over manual auditing: it operates on the full universe of contracts, is reproducible, and does not require confidential information. Future linkage with SISMED unit prices and item-level disaggregation would likely improve the sensitivity and interpretability of the detector (13,15). The sustained increase in alert rates from 2021 to 2025 supports evaluation of this approach for routine audit workflows.

4.3 Smart contracts applied to pharmaceutical payment workflows: interpretation of a complementary proof of concept

The proof of concept supports only a narrow technical claim: conditional digital state transitions can be encoded and executed in a prototype payment workflow under testnet conditions. It does not demonstrate real-world reduction of payment times, lower transaction costs, or institutional deployability. The reported times are not measurements of the full payment process, which also depends on transport, packaging, storage, institutional validation, treasury procedures, contracting rules, and other bottlenecks not modeled here. A defensible operational evaluation would require prospective integration with public hospital information systems, real transaction traces, explicit counterfactual design, and independent technical and regulatory review. Large-scale implementation would additionally require public-treasury integration, legal validation, cybersecurity controls, and external smart-contract auditing.

4.4 Contribution to health data governance

Colombia has public data sources that can support pharmaceutical surveillance, but it lacks analytical infrastructure that reliably converts those data into actionable evidence. BigLoI illustrates that such infrastructure can be built using open technologies, reproducible workflows, and auditable queries over a contractual corpus.

4.5 Limitations

SECOP-II data quality. Buyer and supplier variables include null, generic, or inconsistent values in a meaningful share of contracts in local sample files, which constrains some network analyses. This reflects source-data quality issues in SECOP-II. The closed 2020-2025 cohort included 162,271 pharmaceutical contracts, while the recovered local PostgreSQL source currently indexes 165,549 pharmaceutical records including live 2026 monitoring rows; both reflect the subset with the highest completeness available to the platform.

SISMED temporal mismatch and granularity. The openly available SISMED source covers 2017-2019, whereas the main PostgreSQL procurement cohort analyzed here covers 2020 to 2025. Accordingly, SISMED was not used as a contemporaneous contract-by-contract comparator but as contextual market reference. In addition, the SECOP-II source does not provide a fully structured breakdown of line items and unit prices for all contracts, limiting direct validation of unit-price overpricing.

Blockchain test-network scope. Validation of the payment workflow was conducted on Sepolia exclusively as a functional test of transitions between digital states. Prototype times are referential and do not correspond to real-world measurements of the full operational process. The scenario analysis does not constitute an economic evaluation, and the prototype was not tested against live institutional payment data. Implementation on mainnet would require independent smart-contract auditing, regulatory validation, coordination with public-treasury systems, cybersecurity review, and ideally integration with hospital systems to capture more granular operational events.

Selection bias in pharmaceutical filtering. Use of 19 inclusion keywords may generate false negatives and false positives. A dedicated taxonomy-validation study would be needed to quantify classification performance.

Interpretive scope. The observational design identifies statistical patterns but does not establish causal relationships or measure clinical impact. Interpreting flagged contracts as evidence of corruption would require additional causal methods and involvement of competent institutions.

4.6 Implications for public policy

These findings contribute to three policy discussions: health-system reform, procurement transparency, and health-data governance. The principal contribution of the study is methodological and reproducible: an open infrastructure capable of transforming dispersed public data into auditable evidence. Concentration and recurrence indicators may support audit prioritization, while uneven SECOP-II data quality deserves institutional attention from Colombia Compra Eficiente.

5. Conclusions

This study describes BigLoI, a national-scale computational pharmaceutical-surveillance platform that monitors SECOP-II contracts from 2015 onward and empirically analyzes a closed indexed cohort of 162,271 pharmaceutical contracts from 2020 to 2025. Three main findings emerge.

1. **Market concentration.** The top 3% of suppliers concentrated 85.87% of total contracted value, the top 10 suppliers concentrated 28.77%, and the HHI was 119.26 on the 0-10,000 scale.
2. **Atypical statistical patterns.** A total of 685 contracts triggered Z-score alerts with absolute Z-score greater than or equal to 1.5 sigma, antibiotics showed Zmax of 8.32, and the alert rate increased from 0.31% in 2021 to 1.38% in 2025.
3. **Reproducible infrastructure.** Integration of relational data storage, reproducible analytics, and a public observatory makes it possible to convert dispersed public sources into an auditable surveillance instrument for pharmaceutical procurement.

As a complementary result, the Sepolia prototype shows that conditional payment logic can be represented and executed in a controlled testnet environment. Any inference about reduced payment delays or financial savings remains hypothetical and dependent on assumptions that were not empirically validated in this study. This line of work therefore remains exploratory and still requires operational, regulatory, technical, and logistical validation.

Overall, BigLoI provides evidence that an open and reproducible surveillance infrastructure can generate useful findings on market concentration, statistical anomalies, and auditable evidence

access from public administrative data on Colombian pharmaceutical procurement.

Author contributions

Andres Soto: study conception and design; platform development; data acquisition, processing, and analysis; manuscript writing.

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Competing interests

The author declares no competing interests.

Data and code availability

- **Article reproducibility package:** Archived in Zenodo with DOI <https://doi.org/10.5281/zenodo.19074137> and available in GitHub at <https://github.com/zswamtech/BigLoI-PLOS-ONE-paper>.
 - **Extended BigLoI platform source code:** Available in GitHub at <https://github.com/zswamtech/BigLoI-PMV>.
 - **SECOP-II, INVIMA, and SISMED data:** Available through datos.gov.co.
 - **Author ORCID:** <https://orcid.org/0009-0004-8001-5372>.
 - **BigLoI National Medicines Observatory:** The public observatory is undergoing editorial finalization, and its URL will be provided once stable production deployment is restored.
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Figure legends

Figure 1. Annual evolution of public pharmaceutical procurement monitored in SECOP-II (closed cohort, 2020-2025): number of contracts and total contracted value.

Figure 2. Geographic distribution of total value in public pharmaceutical procurement monitored in SECOP-II by department (closed cohort, 2020-2025).

Figure 3. Distribution of statistical contract-value anomalies detected by the Z-score engine, by therapeutic category and year (2020-2025).

Figure 4. Market concentration in public pharmaceutical procurement monitored in SECOP-II (closed cohort, 2020-2025): Lorenz curve and top 10 suppliers.

Figure 5. Conceptual scheme of the pharmaceutical payment workflow: current institutional flow versus digital states of the smart-contract prototype.

Figure 6. Simplified technical architecture of the BigLoI platform: seven layers for processing public pharmaceutical data.

Table 1. Summary of the corpus of public pharmaceutical procurement monitored in SECOP-II.

Table 2. Technical architecture of the BigLoI platform for surveillance of public pharmaceutical procurement.

Appendices

Appendix 1. SECOP-II pharmaceutical filtering taxonomy

Inclusion terms (19): medicamento, farmaceutico, farmacia, antibiotico, vacuna, insulina, quimioterapia, dispositivos medicos, insumo medico, suministro medico, biologico, antiviral, analgesico, anestesia, oncologico, hemodialisis, hormonal, anticoagulante, inmunosupresor.

Exclusion terms (5): obra civil, construccion, mantenimiento, tecnologia de informacion, mobiliario.

Appendix 2. Simplified database schema

Main tables: observatorio_nacional.contratos_secop (956,157 total records in the recovered local snapshot; 165,549 pharmaceutical records currently indexed, of which 162,271 belong to the closed 2020-2025 cohort); observatorio_nacional.sismed_precios_referencia (44,038 records; 1,759 unique ATC codes; years 2017-2019); observatorio_nacional.invima_medicamentos (9,838 current records); puntos_autorizados (101 hospitals); medicamentos (approximately 10,000 form records); sim_hospitales_100 (100 simulated hospitals).

Appendix 3. Smart-contract protocol

Smart contracts: InvoiceRegistry.sol, InvoiceNFT.sol, and PaymentEscrow.sol; network: Ethereum Sepolia testnet; oracle layer: Chainlink CRE; invoice NFTs used for invoice traceability; escrow module used for conditional payment release.